Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | San Francisco, CA

PROFESSIONAL SUMMARY

AI Engineer with 9 years of production Python + TypeScript engineering and deep expertise in **LangChain/LangGraph**. Built **AgentProbe** -- an open-source LLM eval + agent observability toolkit on LangChain/LangGraph -- with a 5-metric LLM-as-judge evaluation engine, ChromaDB RAG pipeline, CI/CD quality gates, and React observability dashboard. Shipped 20+ production modules (100K+ MAU, 1.6M+ records) using the same evaluation-first mindset. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across 190+ countries; applied Isolation Forest to automate data-quality pipelines across 3 research projects -- surfaced anomalies manual review missed.
- Designed Tableau, ggplot2, and Quarto dashboards translating ML outputs into faculty decisions -- 40% comprehension improvement across two cohorts.

Entrata
Jan 2017 -- Aug 2024

Software Engineer

- Shipped 20+ full-stack microservices (Python, TypeScript, PostgreSQL, RabbitMQ, FastAPI) across 120+ Agile sprints for **100K+ monthly active users** -- 40% deployment time reduction; 50% execution time cut via Docker containerization.
- Applied Python, PyTorch, and OpenAI API to automate AI pipelines across **1.6M financial records** -- production LLM systems with output validation and quality gates on regulated data.
- Improved query performance 30% across 200+ SQL queries; achieved 100% PHI compliance on AWS S3 migration with Libsodium-encrypted pipelines -- zero security incidents; built affordable housing compliance modules (Section 8/HCV, LIHTC, HUD) - income eligibility verification, AMI threshold calculations, and HUD regulatory reporting across 100K+ unit portfolios.

PROJECTS

AgentProbe: LLM Eval + Agent Observability Toolkit
Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built on **LangChain + LangGraph** -- 3-agent runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider model routing (Anthropic, OpenAI, Groq, Ollama) -- deployed as 15-endpoint FastAPI backend.
- Automated quality gates with a **5-metric LLM-as-judge evaluation engine** (factual accuracy, hallucination, retrieval, completeness, compliance) + CI/CD integration; React dashboard tracking per-run metrics, regression drift, and retrieval quality across concurrent agent test suites.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building an NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and semantic retrieval -- **HIPAA-compliant RBAC** across structured healthcare records.

PropLens: Lease Intelligence & Compliance AI
In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs · React

- Building agent-based RAG pipeline over lease PDFs using LangGraph + ChromaDB -- LLM-as-judge scoring and anomaly detection across 1,000+ tenant records; extends AgentProbe eval patterns to a domain-specific compliance use case.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Jawaharlal Nehru Technological University, Anantapur — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, LLM Evaluation, LLM-as-Judge, Eval Frameworks, Agent Observability, Multi-Agent Orchestration, ChromaDB, RAG Pipelines, Embeddings, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, Hugging Face, FastAPI

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, BERT, spaCy, NumPy, Pandas, Tableau, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL